



Abhishek Kumar

abhithakur588@gmail.com | +918090413591 | LinkedIn | Github | Portfolio

PROFILE SUMMARY

I am a seasoned data science professional with more than 6 years of experience. My expertise spans designing and implementing solutions for real-life challenges in the retail, energy, and finance industries using ML/DL/GenAI concepts, big data, and cloud-based technologies. I am adept at writing production-ready code. Good at problem-solving, solution implementation, and delivering scalable solutions aligned with business goals.

EDUCATION

IIT (ISM) DHANBAD

M.TECH IN COMPUTER SCIENCE
2018 - 2020 | Dhanbad, Jharkhand
CGPA: 8.87 / 10.0

UNIVERSITY OF ALLAHABAD

B.TECH IN COMPUTER SCIENCE
2014 - 2018 | 74.1% | Allahabad, UP

JNV SAHARSA

INTERMEDIATE CBSE(XII)
2011 - 2013 | 90 % | Saharsa, Bihar

SKILLS

PROGRAMMING

Python • Java • SQL • C++

DATA SCIENCE

Machine Learning • Deep Learning
NLP • LLM • Prompt • GenAI
Docker • Podman • Kubernetes
GCP • TensorFlow • Torchserve
Hadoop • Hive • PySpark
MLflow • BashScript • Airflow
Numpy • Pandas • sklearn
Open CV • Pytorch • NLTK
BigQuery • Kafka • vLLM
Ray • LangChain • LangGraph

PRODUCTIVE TOOLS

Copilot • Cursor • Git
VS Code • Anaconda • Jupyter
Excel • Tableau • Jira • Claude

CODING PROFILES

HackerEarth • LeetCode
CodeChef • Kaggle

CERTIFICATIONS

Data Scientist with Python
GCP Big Data & ML Fundamentals
Core Java (J2SE)

EXPERIENCES (6 YEARS)

WALMART | SENIOR DATA SCIENTIST (MLE)

Apr 25 - Present | Bangalore India

- Engineered a real-time **Item setup** pipeline for product **title, description, and attributes** recommendations using LLMs and agents, serves **500k items/day**, integrated as **API** and other **microservices** for sync, **Kafka** for async processing.
- Worked on **Gemini, Gemma, Mistral** Multimodality finetuning using **PEFT** for **Attribute Extraction and Content Generation**, designed high throughput **Inferencing** pipeline using **vllm**, and **triton** server which serves **10M req/day** results in high content quality score and coverage up to **60%** and recall of **80%**.
- Developed a cleanup cum **product attribute extraction** model using **NLP**, **regex**, **fuzzy**, and **embeddings** integrated with **faiss** vector db for **semantic** search on **1 mn+** vectors with custom index, deployed as **Fast API** in **K8** cluster.
- Worked on **LLM as a judge framework** for auto AE validation with **reasoning**.
- Designed **Grounding search** based content extraction for **1p** and **3p** suppliers.
- Tech-**GenAI, GCP, NLP, LLMs, Pyspark, Airflow, Kafka, Kubernetes, vLLM, Ray**.

SLB (SCHLUMBERGER) | MACHINE LEARNING ENGINEER II

July 22 - Apr 25 | Pune, India

- Developed **vision-based algorithms** for productivity (workstation and tools utilization) and health and **safety non-compliance (mechanical lifting, PPE kit, zone intrusion)** in industrial floor settings. **Achieved 85%** compliance accuracy.
- Trained and fine tuned **DL models** on custom image datasets of **persons, hats, gloves, and tools** for **object detection, segmentation, and classification**.
- Engineered **model service architecture** using **TorchServe**, increased **fps 2 to 4**.
- Streamlined deployment in different locations based on a **custom developed CI/CD module** using **Docker, Azure Pipelines, Podman, GCR, model registry and other GCP services**. Which reduced use-case deployment time by **75%**.
- Revamped architecture through **scheduling, logging, Pub/Sub integration, code modularization**, testing, monitoring, dashboards, and **MLops integration**.
- Productionized several **real time video analytics** based usecases. Integrated **model and data versioning**, Production **monitoring** using internal **Flask based webapp** and **PowerBI** dashboards. Infrastructure on-boarding, **sustenance**.
- Tech-**Python, OpenCV, Torch, Tensorflow, YOLO, Torchserve, Image processing, GCP(Bigquery, PubSub, Vertex AI, GCR), Docker, GenAI, SAM**.

NATIONAL PAYMENTS CORP. OF INDIA | ASSOCIATE DATA SCIENCE

Nov 2020 - July 2022 | Hyderabad, India

- Responsibility includes **fraud Model building**, **feature engineering**, **Exploratory data Analysis**, **Model training and testing**, **data pipelines script generation**, **scheduling**, **service validation**, **user profiling**, **user segmentation**.
- Built several ML models for **P2P(Person to Person)**, **P2M(Person to merchant)** **fraud detection** which serves almost **25cr UPI transactions** per day. Built **vpa (text) clustering** model using **NLP library**. Reduced fraud by **60%** in ecosystem.
- Contributed in Generation of more than **200 features** for every customer on **UPI platform**, **EDA, Visualization**, automating training and testing of ml models.

PUBLICATIONS & ACHIEVEMENTS

- PATENT** publication on Innovation title **Video analytics for industrial floor setting**. Reference **PCT/US2024/63/485,939**
- Received **Bravo & Excellence Award** for my contribution at **Walmart**
- Received **PBS CEO & Bronze Award** for Digital Factory at **SLB**
- Received **Circle of Excellence Award** for my contribution at **NPCI**